KPI Definitions and Data-Quality Documentation

Week 4: Consumer Complaint Trends and Response Intelligence Platform

Item	Detail
Programme	AnalystLab Africa | Experience Lab Internship Programme
Track	Data Analytics
Week / project	Week 4: Consumer Complaint Trends and Response Intelligence Platform
Author	Godfrey Miracle
Dataset	CFPB consumer complaints on Bank of America, 1 May 2017 to 28 August 2023 (62,516 records, 12 columns)
Tools	Power Query, Power BI (DAX) and MySQL 8.0 (independent cross-check)
Document date	9 November 2026

1. Purpose

This document defines every KPI shown on the Power BI dashboard, records how each field was prepared, logs every data-quality issue and the decision taken and summarises the validation that checks the dashboard figures independently. It supports the Data Analytics track deliverables: KPI definitions and calculations and data-quality documentation.

Reading the figures. A complaint is an issue reported by a consumer. It does not establish misconduct or prove that the company violated any rule. Recorded company responses are not proof of satisfactory resolution. All findings below describe what the records show.

2. KPI definitions

Values are for the full dataset with no slicer selected and every value was reproduced in SQL (see section 5), except the May to December 2018 comparison in KPI 12, which was calculated separately.

#	KPI	Definition and rule	Power BI measure	Verified value
1	Total complaints	Count of complaint records. Each row is one complaint; Complaint ID is unique.	Total Complaints = COUNTROWS(Complaints_Clean)	62,516
2	Annual complaint volume	Complaints grouped by the year of Date submitted (the date the CFPB received the complaint).	Total Complaints by Calendar[Year]	2017: 5,394 (May to Dec only) 2018: 7,872 2019: 7,075 2020: 8,942 2021: 11,149 2022: 12,953 2023: 9,131 (to 28 Aug only)

#	KPI	Definition and rule	Power BI measure	Verified value
3	Monthly complaint volume	Complaints grouped by calendar month of Date submitted.	Total Complaints by Calendar[Year-Month]	76 months Highest: Jul 2023 (1,749) Lowest: Feb 2019 (356) Aug 2023 is partial (to 28 Aug)
4	Complaints by financial product	Complaints grouped by Product. A short display label (Product Short) is used on charts; the original Product is unchanged.	Total Complaints by Product Short	9 products Checking / savings 24,814 Credit / prepaid card 16,197 Credit reporting 7,710
5	Complaints by reported issue	Complaints grouped by Issue, as reported by the consumer. Not a verified finding.	Total Complaints by Issue (Top N on Page 2)	76 issues Managing an account 15,109 (24.17%) Top 10 issues = 65.9% of complaints
6	Complaints by sub-product and sub-issue	Complaints grouped by Sub-product and Sub-issue. Blank values are labelled Not specified.	Total Complaints by Sub-product / Sub-issue	46 sub-products (7 blank) 158 sub-issues (10,858 blank, 17.37%) Top sub-product: Checking account 20,768
7	Percentage contribution of each product	Product complaints divided by complaints in the current filter selection (ALLSELECTED), so shares still add to 100% when a slicer is used.	Share of Complaints % = DIVIDE([Total Complaints], CALCULATE([Total Complaints], ALLSELECTED(Product)))	Checking / savings 39.69% Credit / prepaid card 25.91% Credit reporting 12.33% Mortgage 10.56% Money transfer / service 5.52% Debt collection 4.38% Vehicle loan / lease 1.01% Payday / personal loan 0.53% Student loan 0.06%

#	KPI	Definition and rule	Power BI measure	Verified value
8	Company response category distribution	Complaints grouped by Company response to consumer, as recorded. A category is not proof of resolution.	Total Complaints by Company response to consumer	Closed with explanation 41,044 (65.65%) Closed with monetary relief 14,697 (23.51%) Closed with non-monetary relief 5,273 (8.43%) In progress 1,494 (2.39%) Closed 8 (0.01%)
9	Timely response percentage	Records marked Yes divided by records marked Yes or No. Records with a blank value (In progress) are excluded from the denominator and reported separately.	Timely Response Rate = DIVIDE([Timely Responses (Yes)], [Valid Timely Records])	96.06% (58,619 / 61,022) Untimely (No): 2,403 (3.94%) Excluded: 1,494 In progress
10	Complaints marked In progress	Complaints whose Company response to consumer is In progress. These have no timely-response value and no public response.	In Progress Complaints	1,494
11	Average and median forwarding interval	Days between Date submitted (CFPB received the complaint) and Date received (CFPB sent it to the company). It is the CFPB forwarding interval, not the company's response or resolution time.	Avg / Median Forwarding Interval = AVERAGE / MEDIAN of Forwarding Interval (Days)	Average: 1.22 days Median: 0 days Same-day: 79.4% Maximum: 275 days Over 30 days: 435 (0.70%)
12	Year-over-year change	(Complaints this year minus prior year) divided by prior year, using SAMEPERIODLASTYEAR on the Calendar table.	YoY Change % = DIVIDE([Total Complaints] - [Prior Year Complaints], [Prior Year Complaints])	2018: +45.94%* 2019: -10.12% 2020: +26.39% 2021: +24.68% 2022: +16.18% 2023: -29.51%* * Partial-year comparisons. Like-for-like: Jan to 28 Aug 2023 vs 2022 = +11.3%; May to Dec 2018 vs 2017 = -8.9%

3. Prepared and calculated fields

Field	Rule	Used for
Year, Month, Quarter, Year-Month	Derived from Date submitted. Year-Month is the first day of the month and is used on the monthly trend axis.	Pages 1 and 4
Forwarding Interval (Days)	Date received minus Date submitted, in whole days. Stored as a whole number; no negative values exist.	KPIs 11; Page 4
Interval Band	0 days, 1-3 days, 4-7 days, 8-30 days, Over 30 days, with an Interval Band Order helper column so the bands sort correctly.	Page 4
Product Short	Display label only (for example Credit / prepaid card). The original Product column is kept.	Pages 1, 2, 4
Timely Status	Yes or No when the source has a value; In progress when the source is blank. Blank is never treated as No.	KPIs 9, 10; Page 3
Public Response Status	Source value when present; Pending (In progress) when the complaint is In progress; Not recorded (closed) when the complaint is closed but the public response is blank.	Page 3
Public Response Short	Shorter labels for the six recorded public response texts, for chart readability.	Page 3
Calendar table	Date table (1 Jan 2017 to 31 Dec 2023), marked as the date table, related to Date submitted (one-to-many, single direction).	KPIs 2, 3, 12

4. Data-quality log

Each issue was investigated before analysis. Nothing was deleted; unusual values were kept and flagged.

#	Issue	Finding	Decision	Impact
1	Unique key	Complaint ID has 62,516 distinct values for 62,516 rows (0 duplicates).	Used as the primary key; stored as text because it is an identifier.	None
2	Duplicate records	No fully duplicated rows.	No rows removed.	None
3	Blank Sub-product	7 records (0.01%).	Labelled Not specified.	Negligible
4	Blank Sub-issue	10,858 records (17.37%).	Labelled Not specified; reported separately in sub-issue analysis.	Sub-issue charts show a large Not specified group

#	Issue	Finding	Decision	Impact
5	Blank Timely response?	1,494 records. All are In progress.	Timely Status = In progress. Excluded from the timely-response denominator and reported separately.	Timely rate = 58,619 / 61,022 = 96.06%
6	Blank Company public response	2,175 records: 1,494 In progress (no response expected yet) plus 681 closed complaints with no recorded public response.	Split into Pending (In progress) and Not recorded (closed).	The reason for the 681 is not stated in the data
7	Date validity	All dates valid. Both date columns run from 1 May 2017 to 28 Aug 2023. No forwarding interval is negative.	No changes.	None
8	Unusual forwarding intervals	Median 0 days; maximum 275 days; 435 records (0.70%) are over 30 days.	Kept and flagged, not removed. Shown in the Long intervals table on Page 4.	Averages are pulled up by a few long intervals; the median is reported alongside
9	Submitted via	Web Referral (90 records) alongside Referral (10,766).	Merged into Referral (10,856). This is a judgement call because the dataset does not define Web Referral.	Referral count includes 90 merged records
10	Company response: Closed	8 records labelled only Closed, next to three more specific Closed categories. All 8 are marked not timely.	Kept exactly as recorded and noted as undefined.	Meaning cannot be confirmed from the data
11	Partial periods	2017 starts on 1 May (8 months). 2023 ends on 28 Aug and activity tails off after mid-August (1 to 5 complaints a day in the last week against 40 to 60 earlier).	Kept. Labelled on the dashboard; like-for-like comparisons calculated for YoY.	YoY for 2018 and 2023 is not like-for-like
12	Long category names	Several Product, Issue and public response values exceed 70 characters.	Display labels added (Product Short, Public Response Short); source columns untouched.	None on counts
13	Original data preserved	Complaints_Raw is kept unedited in Power Query and is not loaded to the report. The SQL raw table is never modified.	All cleaning is done in Complaints_Clean (Power Query) and the SQL view complaints_clean.	Fully reproducible

5. Validation summary

The cleaning and every KPI were rebuilt in MySQL (script Week4_Validation.sql) and compared with the dashboard. The script ends with 25 automatic checks; all 25 returned PASS. A selection is shown below.

Check	Verified value	Result
Total complaints	62,516	PASS
Distinct Complaint IDs / negative intervals	62,516 / 0	PASS
Date range	2017-05-01 to 2023-08-28	PASS
Timely Yes / No / In progress	58,619 / 2,403 / 1,494	PASS
Timely response rate	96.06%	PASS
Average forwarding interval / maximum	1.2249 / 275	PASS
Complaints over 30 days	435	PASS
Complaints in 2017 / 2022 / 2023	5,394 / 12,953 / 9,131	PASS
Checking / savings and Mortgage complaints	24,814 / 6,601	PASS
Managing an account complaints	15,109	PASS
Pending and Not recorded public responses	1,494 / 681	PASS
Referral after merging Web Referral	10,856	PASS
July 2023 (peak month)	1,749	PASS

6. Assumptions and limitations

No narratives. The data contains no complaint narratives, so no sentiment or cause analysis is possible.
Forwarding interval. The interval between Date submitted and Date received is the CFPB forwarding interval. It is not the company's response time or resolution time.
Recorded indicators. Timely response and company response categories are as recorded. They do not show whether a complaint was resolved to the consumer's satisfaction.
Partial years. 2017 covers 8 months and 2023 covers about 8 months, so annual totals and year-over-year changes for 2018 and 2023 are not like-for-like.
Unexplained peak. The July 2023 peak (1,749 complaints) is concentrated in checking and savings and credit card products, mid-month, mainly under Managing an account. The data does not explain the cause.
Issue taxonomy. Most issue categories belong to a single product, so comparing issue counts across products mostly reflects each product's own issue list.
Status at extraction. Complaints In progress reflect their status when the data was extracted and may have changed since.
Scope. The dataset covers one company only. It cannot show how Bank of America compares with other institutions, why complaint volumes changed, or whether individual complaints were justified.
Judgement calls. Web Referral merged into Referral and the Not specified labels are documented judgement calls. They do not change any KPI total.